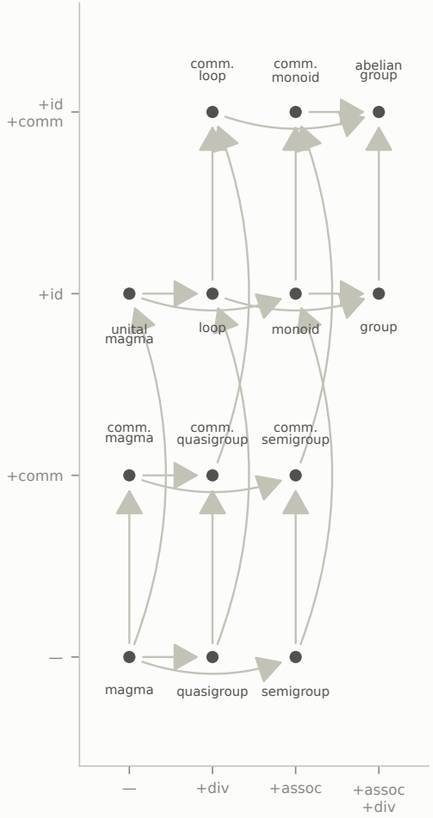
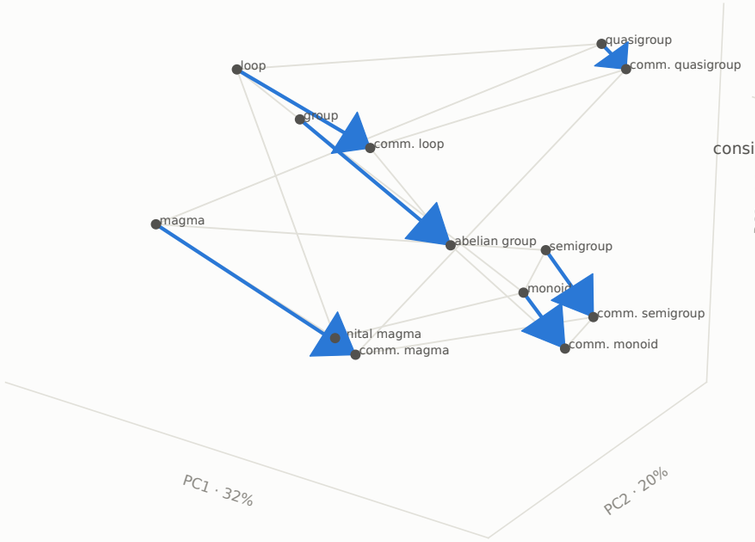
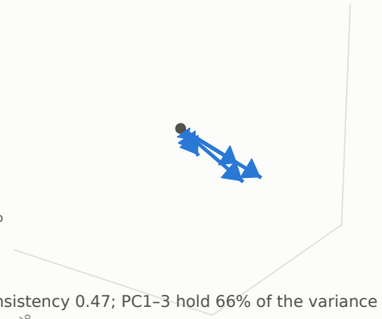
1 · Structures are sets of axioms



2 · Adding an axiom is an arrow + commutativity in Qwen3.5-2B (top-3 PCs)

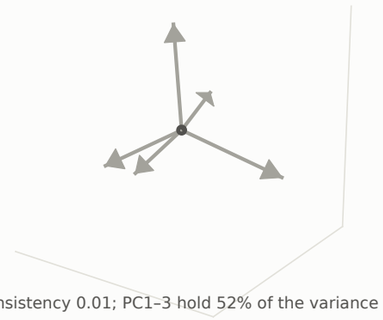


3 · Line the arrows up model



consistency 0.47; PC1–3 hold 66% of the variance

random-init twin

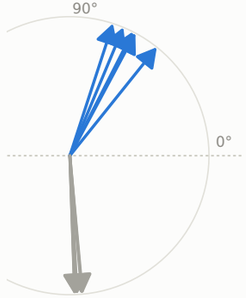


consistency 0.01; PC1–3 hold 52% of the variance

4 · Measure in the full space: angle of each arrow to the others' mean

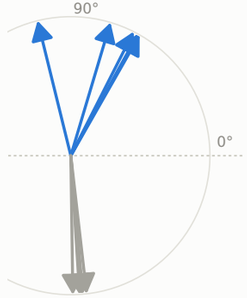
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



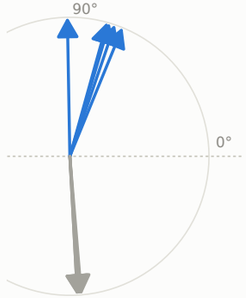
consistency 0.31 (p = 0.021)
twin 0.03

+ identity



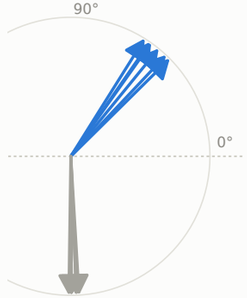
consistency 0.16 (p = 0.091)
twin 0.04

+ divisibility



consistency 0.14 (p = 0.124)
twin 0.03

+ commutativity



consistency 0.47 (p = 0.002)
twin 0.01